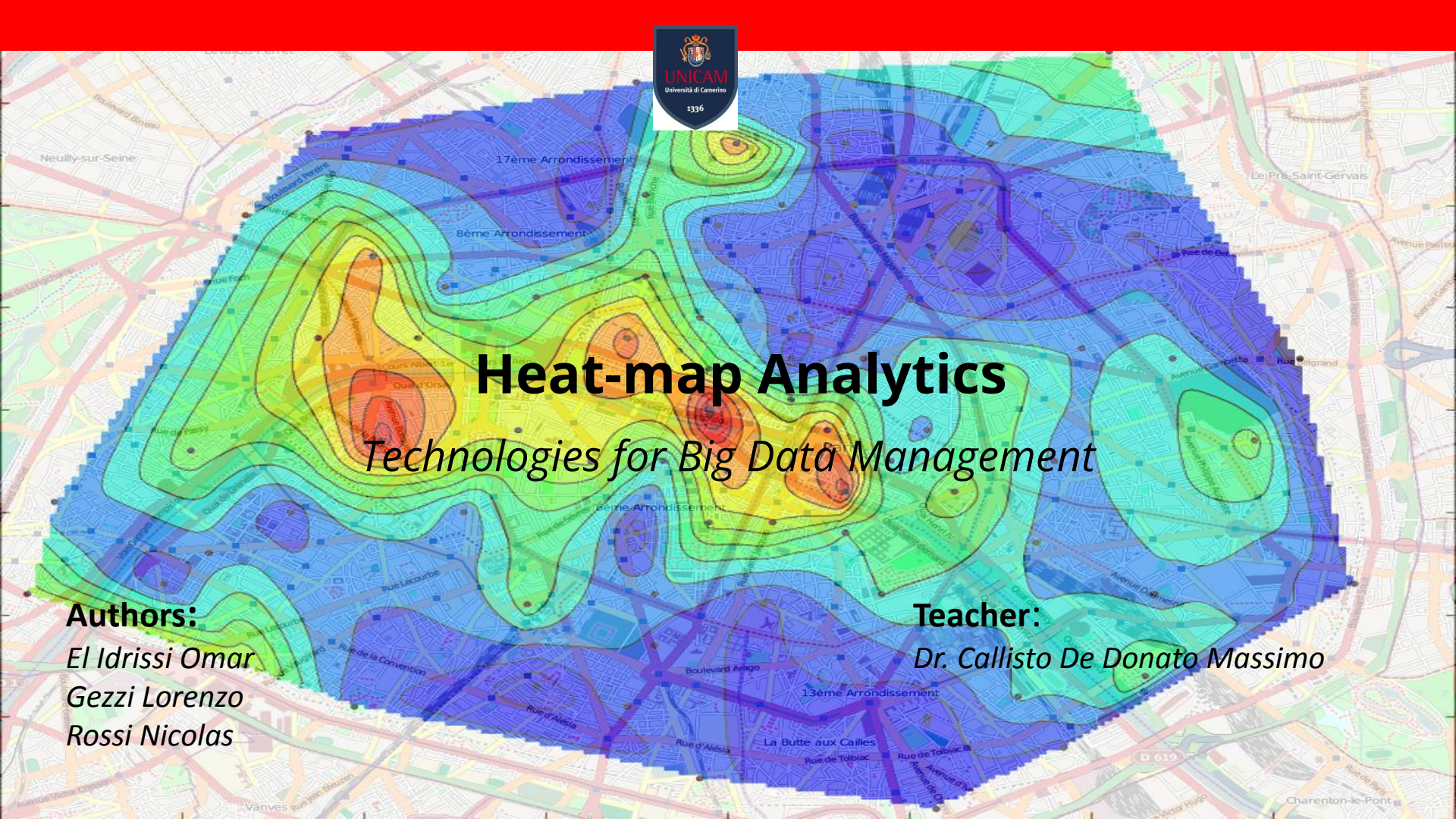




A detailed heat-map of Paris, France, overlaid on a street map. The map uses a color gradient from blue (low density) to red (high density) to represent areas of high activity or population density. The map shows various arrondissements, including the 17ème, 8ème, 13ème, and 16ème. The title 'Heat-map Analytics' is prominently displayed in the center, with the subtitle 'Technologies for Big Data Management' below it.

Heat-map Analytics

Technologies for Big Data Management

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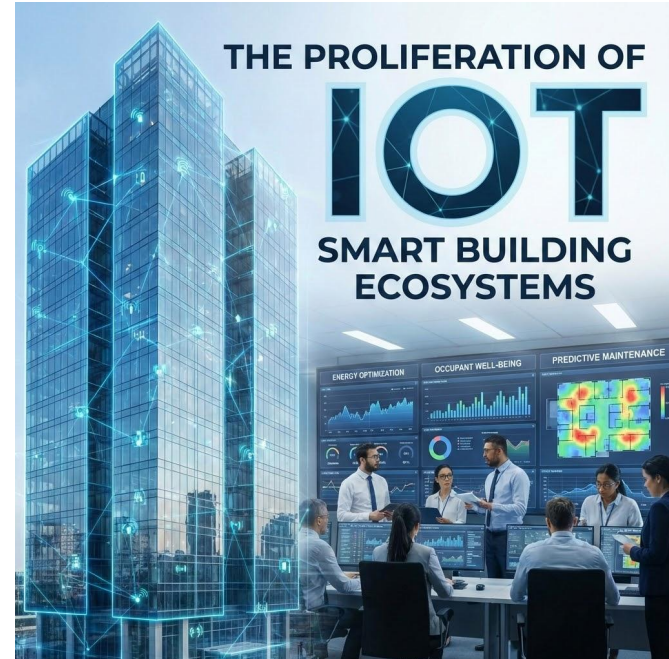
Thingsboard Integration

08

**Conclusions and Future
Developments**

Context - The Rise of Smart Infrastructure

- **The Proliferation of IoT:** Modern buildings are evolving into "intelligent" ecosystems, utilizing a vast sensor network for environmental monitoring.
- **Data-Driven Operational Efficiency:** Telemetry is now essential for optimizing energy, enhancing occupant well-being, and enabling proactive maintenance in large infrastructures.
- **The Big Data Shift in Facilities:** There is a growing industry transition from isolated local automation systems toward integrated Big Data stacks that allow for advanced historical analysis and real-time operational decision-making.



Challenge - Scaling IoT Intelligence

- **Scalability & Memory Pressure:** High-frequency telemetry from hundreds of sensors can quickly overwhelm system memory and processing threads without proper decoupling or throttling.
- **Integration Friction:** Connecting diverse sources requires a highly flexible ingestion layer to avoid rigid, fragile data pipelines.
- **The Feedback Gap:** Analytical insights often remain database-bound; real-time visualization on the IoT control plane requires an automated feedback loop.

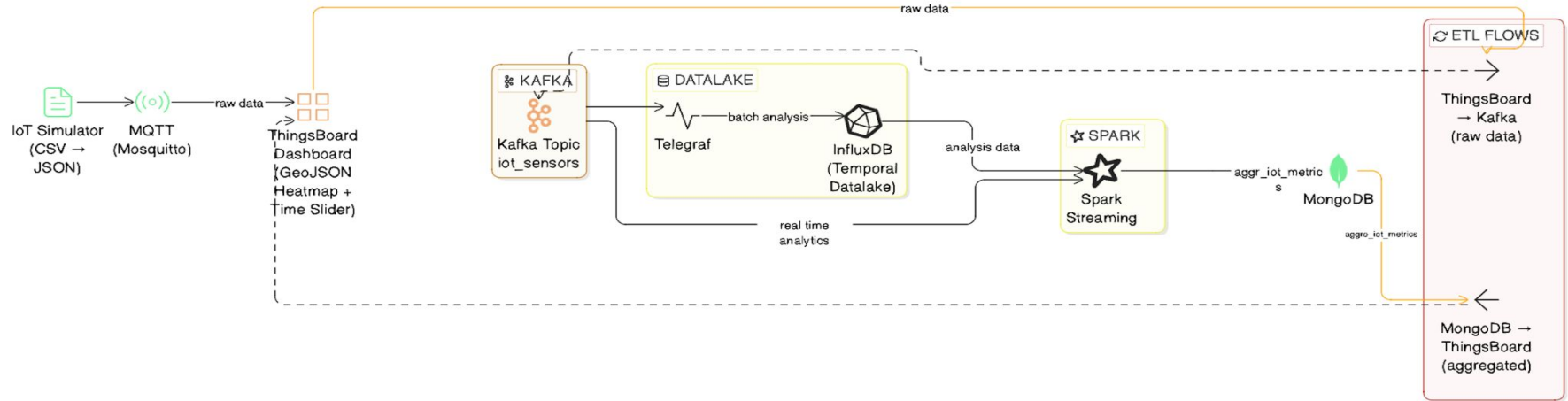




Solution - A Resilient & Integrated Big Data Ecosystem

- **Hybrid Ingestion:** Dual-profile layer supports Polling ETL (consistent) and Rule-Chain/FastAPI (real-time) for flexible data intake.
- **Resilient Buffering:** Apache Kafka ensures high-throughput, decoupling data collection from analytics for fault tolerance.
- **Multi-Layered Analytics:** Apache Spark uses dual processing: Streaming for real-time telemetry (Avg, Min, Max); Batch for deep historical reporting via Ofelia scheduler.
- **Closing the Loop:** Pipeline returns processed insights to ThingsBoard for immediate monitoring and alerts.

The Architecture



- Decoupled IoT pipeline leverages Kafka.
- Bridges real-time Spark streaming and InfluxDB batch analysis for comprehensive telemetry processing.
- System closes the operational loop by persisting insights in MongoDB.
- Feeds aggregated analytics back to the ThingsBoard dashboard.



Data Simulation & Infrastructure Provisioning

IoT Telemetry Simulation:

- A Python-based IoT Simulator converts historical CSV data into standardized JSON payloads.
- Data is transmitted via MQTT, mimicking physical gateways for a real-world building environment.

Automated ThingsBoard Provisioning (Initialization Script):

- **Entity Mapping:** Programmatically creates Building, Floor, and Room assets in ThingsBoard.
- **Spatial Intelligence:** Generates and uploads a GeoJSON Grid to define room boundaries.
- **Device Synchronization:** Automatically provisions and assigns every mapped sensor to its specific room asset, ensuring telemetry is linked to location before the pipeline starts.





Data Format & Kafka Payload

- Standardized JSON across the pipeline ensures seamless interoperability between Kafka producers and Spark consumers.
- Raw telemetry is enriched with spatial metadata upon ingestion.
- Payload Specifications:
 - Timestamp is Unix seconds.
 - Sensor Metadata includes ID and type.
 - Telemetry values are floating-point numbers for precise Spark aggregations.

```
{  
  "timestamp": 1741104000,  
  "building": "POLOA",  
  "floor": "4",  
  "room": "417",  
  "sensor_type": "temperature",  
  "sensor_id": 1,  
  "value": 22.5,  
  "device": "temperature_1"  
}
```



Real-Time Processing with Spark Streaming

- **Continuous Analytical Engine:** Uses Apache Spark Structured Streaming for low-latency processing of enriched data from the Kafka `iot_sensors` topic.
- **Stateful Windowed Aggregations:** Telemetry is processed via sliding time windows to capture dynamic trends (**default set at 10 minutes**).
- **Real-Time Statistical Metrics:** Computes average, min/max, standard deviation, and variance for each window/device in real-time.
- **Persistence & High Availability:** Aggregates are stored in **MongoDB** for high-performance dashboard querying. Checkpoints ensure streaming state is preserved for fault recovery.





Historical Insights & Batch Processing

- **Complementary Analytical Layer:** Streaming handles alerts; Spark Batch performs historical computation on raw InfluxDB data.
- **Orchestrated Scheduling with Ofelia:** Docker-native Ofelia schedules automated spark-submit jobs during off-peak hours to minimize load.
- **Deep Temporal Analysis:**
 - **Daily Comfort Reports:** 24-hour telemetry aggregated for building performance and comfort evaluation.
 - **Weekly Infrastructure Health:** Analysis of sensor drift, battery, and connectivity to detect hardware failures/maintenance needs.
- **Persistence of Long-Term Trends:** Final reports are stored in MongoDB for structured history and facility management dashboards.



influxdb



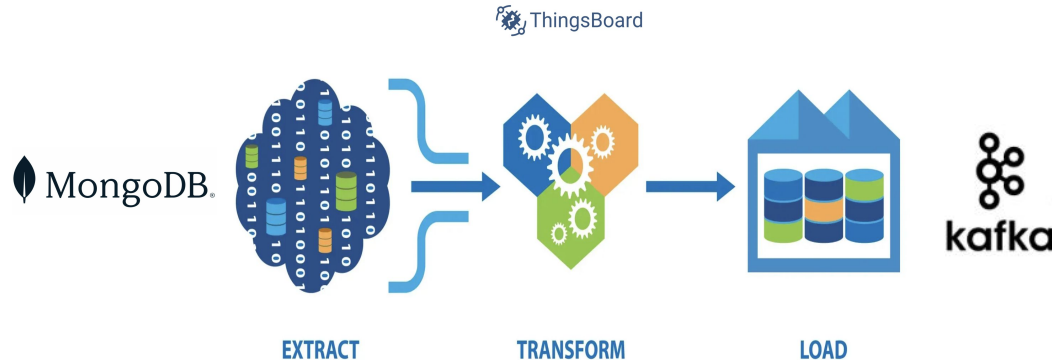
ThingsBoard

ThingsBoard is an open-source IoT platform designed for **data collection, processing, visualization, and device management**. In our architecture, it acts as both the **primary data source** for raw telemetry and the **final visualization layer** for processed analytics.

- **Multi-Protocol Ingestion:** Supports high-performance data telemetry via **MQTT and HTTP**. Our system utilizes the **MQTT Gateway API** to simulate real-time sensor streams with customized intervals.
- **Device Management:** Provides a robust entity model to define **Devices and Assets**. We leverage **Server-Side Attributes** to store spatial metadata such as Building, Floor, and Room.
- **Virtual Device Provisioning:** Allows for the dynamic creation of **Virtual Devices** to represent computed metrics. Our ETL Loader automatically provisions these devices and maps aggregated statistics from MongoDB back to the dashboard.

ThingsBoard Integration & The ETL Ecosystem

- **Data Ingestion (To Kafka):** ThingsBoard is the primary telemetry source. The ETL layer extracts and enriches raw data with spatial metadata (Building, Floor, Room) before forwarding the enriched stream to Kafka.
- **Feedback Loop (From MongoDB):** To close the operational loop, a dedicated ETL monitors MongoDB for Spark-calculated insights and pushes them back to ThingsBoard using APIs to update real-time dashboards.





Adaptive Ingestion: Dual-Strategy Approach

The architecture offers two interchangeable ingestion profiles for balancing consistency and latency:

- **Approach A: Polling Profile (Pull-based)**
 - **Mechanism:** Scheduled ETL script polls ThingsBoard API every 5 seconds.
 - **Feature:** Uses a Watermark (last processed timestamp) for data consistency and duplicate prevention during recovery.
 - **Best for:** Reliable, stateful ingestion where order and "at-least-once" delivery are critical.
- **Approach B: Rule-Chain Profile (Push-based)**
 - **Mechanism:** ThingsBoard automatically forwards telemetry via a custom Rule Chain to a **FastAPI** endpoint.
 - **Feature:** Event-driven bridge transforms and pushes data immediately to Kafka.
 - **Best for:** High-velocity, real-time scenarios requiring minimal sensor-to-processing latency.

Future Works

Infrastructure Resilience (Kafka Optimization):

- Current State: Replication Factor (RF) is 1 (local testing/optimization).
- Target: Scale RF to 3 in a multi-node cluster for high availability, fault tolerance, and data loss prevention.

Security Hardening:

- Implement E2E Encryption (SSL/TLS) for data in transit.
- Introduce Role-Based Access Control for MongoDB and ThingsBoard to secure building metadata.

Unified Batch Visualization:

- Extend the Loop-Closing ETL to push MongoDB Batch results back to ThingsBoard.



Demo Time...

